THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

Time: May 12th, 2021 9:30 AM – 1:30 PM (4 hours)

Answer any **5 (FIVE)** questions.
All questions carry equal marks
Marks for sub-questions (if any) are given.

Note: The answer must be handwritten, including the figures. Please submit one word or pdf document. Compiling photos in one word document is allowed.

Students should **independently** finish the take-home exam.

For the mechanics part, detailed reasoning, equations and steps are necessary.

SECTION A

Q1 A cantilever beam AB is carrying a uniform load of intensity q over part of the span and an intensive force P at point B (see Figure 1).

(a) Derive the equations of the deflection curve of the beam. **(15 marks)**
(b) Determine the deflection at the end of the beam point B. **(5 marks)**

Q2 A slender column ABC is pin-supported at the ends in both directions. It is compressed by an axial load P. Lateral support is provided at the midpoint B in the plane of the Figure 2(a). However, lateral support perpendicular to the plane of the figure is provided only at the ends. The column is constructed of a steel I-beam section $S\,200 \times 34$ (shown in Figure 2(b)) having modulus of elasticity $E = 200GPa$, proportional limit $\sigma_{pl} = 300MPa$, and yield stress $\sigma_{ys} = 450MPa$. The total length of the column is $L = 8m$. The moments of inertia and cross-sectional area for a $S\,200 \times 34$ is $I_1 = 26.9 \times 10^6 mm^4$, $I_2 = 1.78 \times 10^6 mm^4$, and $A = 4360mm^2$.

(a) Determine the maximum compressive loading the vertical bar could resist. **(15 marks)**
(b) Determine the failure mode of the vertical bar. **(5 marks)**

Q3 At a point on the surface of an elliptical exercise machine, the paddle is subjected to biaxial stress with $\sigma_x = 11\ MPa$, and $\sigma_y = -20\ MPa$, as shown in Figure 3.

(a) Find the maximum shear stress and directions (showing stress state) using plane stress transformation and equations. **(10 marks)**
(b) Find the maximum shear stress and directions (showing stress state) using Mohr's Circle. **(10 marks)**

FIGURES

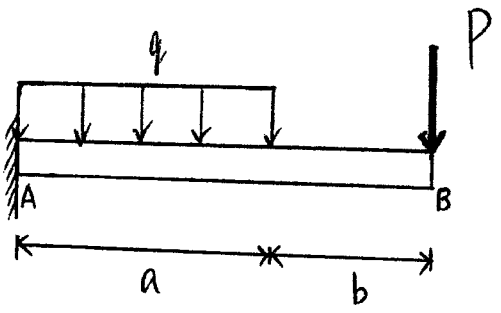
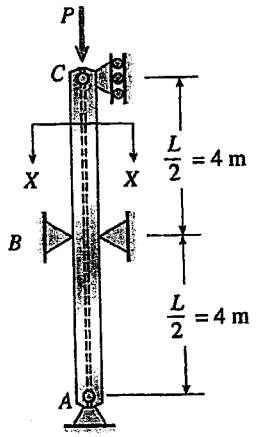


Figure 1



(a)

Figure 2(a)

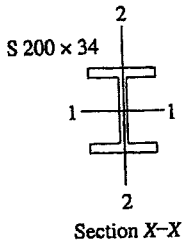


Figure 2(b)

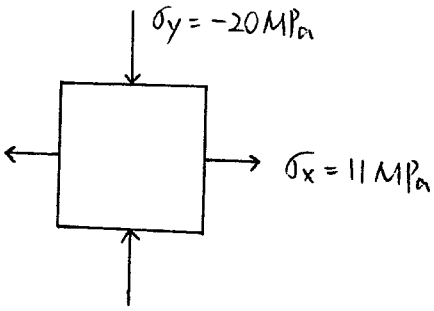


Figure 3

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